

911 Emergency Dispatch System PRD

Problem Statement & Description

A city's 911 emergency dispatch center receives calls for Police, Paramedic, and Fire Department services across 10 fixed locations (L01 through L10). Deciding who responds and manually allocating personnel leads to severe operational inconsistencies—critical emergencies may be dangerously under-staffed, while minor incidents consume unnecessary personnel. Additionally, dispatchers lack real-time visibility into officer workload distribution and incident status updates.

This command-line application enables a 911 operator to act as the central dispatcher. During intake, the operator records four key details: the caller's emergency description and name, a multi-department selection (1 to 3 services chosen simultaneously), a target city location (L01–L10), and a severity rating (1–5). Based on the severity, the system calculates required personnel ($\text{Personnel} = \text{Severity} \times 2$) and assigns lead officer ranks per department. Upon logging, an interactive radio feed simulates real-time unit communication: officers acknowledge case acceptance ("heading to location"), signal arrival ("Reached"), and allow the operator to mark jobs complete once resolved. The application also provides direct terminal access to a police record database for background checks. All logged incidents are organized in a dynamic central queue where active Pending calls remain prioritized at the top and Completed calls accumulate at the bottom, while recording detailed performance metrics under each responding officer's profile.

Interface

Inputs:

- **Main Menu Selection:** User enters an integer from 1 to 5 to navigate system features (1: Log Call & Radio Feed, 2: Access Police Records, 3: Complete Officer Cases, 4: View Queue & Telemetry, 5: Exit).
- **Emergency Intake Details:**
 - Prompt 1:** Caller Name (**text string**) and Emergency Description (**text string**).
 - Prompt 2:** Department Selection (**comma- or space-separated numbers selecting 1 to 3 choices: 1 for Police, 2 for Fire Department, 3 for Paramedics**).
 - Prompt 3:** Location Code (**string chosen from fixed choices L01 through L10**).
 - Prompt 4:** Severity Level (**integer from 1 for Minor to 5 for Critical**).
- **Radio Feed Completion Prompt:** Character input (**Y or N**) entered by the operator after the unit signals arrival on scene.
- **Police Record Search Query:** Text string query (**Civilian Name, SSN, or Location Code**).
- **Call ID for Dispatch:** Unique string identifier (e.g., CALL-001) entered by an officer or operator to mark an active pending call as finished.

Outputs:

- **Main Menu Console:** Displayed menu listing system options alongside active pending call counters.
- **Call Intake Summary:** Confirmation summary displaying Call ID, Caller Name, Incident Description, Selected Departments, Location Code, Severity Rating, Lead Officer Ranks, Personnel Counts, and Initial Call Status.
- **Radio Communication Feed:** Real-time text stream showing unit case acceptance ("heading to location"), status transitions ("En Route" to "On Scene"), and arrival notifications.
- **Police Record Query Display:** Structured output displaying matching civilian profile data, active warrants, traffic citations, and location incident logs.
- **Global Incident Queue & Telemetry Tables:** Dual-table display rendering all calls ordered with **Pending** calls at the top and **Completed** calls at the bottom, alongside an officer performance table showing active vs. completed cases per officer.

Functional Requirements

FR 1: Multi-Department Call Logging

- **Description:** Collects caller metadata, emergency description, chosen services, location, and severity rating.
- **Inputs / Triggers:**
 - Prompts 1–4: Caller Name (text),
 - Emergency Description (text),
 - Department Choices (1–3, comma/space separated),
 - Location Code (L01–L10), Severity Level (1–5).
- **Expected System Action:** Validates inputs, assigns a unique Call ID (e.g., CALL-001), calculates personnel requirements ($\text{Personnel} = \text{Severity} \times 2$) for each chosen department, assigns lead officers, sets call status to **Pending**, and displays an itemized call log confirmation.

FR 2: Operator–Officer Radio Dialogue

- **Description:** Simulates interactive radio feed updates between the dispatch operator and responding lead units immediately following intake.
- **Inputs / Triggers:** Triggered automatically after FR 1 call logging completes.
- **Expected System Action:** System streams unit case acceptance ("Heading to location"), updates unit status to **En Route**, streams scene arrival confirmation ("Reached location"), updates status to **On Scene**, and prompts operator: *"Is the job complete? (Y/N)"*.

FR 3: In-Terminal Police Record Lookup

- **Description:** Provides in-terminal access to civilian criminal background checks, traffic citations, and past location incident logs.

- **Inputs / Triggers:**
 - Main Menu Choice 2 → Enter Search Query (Name, SSN, or Location Code).
- **Expected System Action:** Searches internal database and renders civilian criminal records, active warrants, traffic citations, and location incident history directly on screen.

FR 4: Global Queue & Telemetry Display

- **Description:** Displays all active and completed calls alongside individual officer workload metrics.
- **Inputs / Triggers:** Main Menu Choice 4 or triggered automatically following status updates.
- **Expected System Action:** Renders two formatted tables: (1) Incident queue sorted with **Pending** calls at the top and **Completed** calls at the bottom; (2) Officer telemetry table showing active ongoing vs. completed cases per officer.

FR 5: Officer Case Completion & Queue Reordering

- **Description:** Allows officers or operators to resolve active cases and reorder queue positions.
- **Inputs / Triggers:** Main Menu Choice 3 or Radio Feed completion prompt (Y).
- **Expected System Action:** Validates Call ID, sets call status to **Completed**, credits the assigned lead officer's completed count, and shifts the call to the bottom position in the global queue.

FR 6: Dynamic Severity Escalation

- **Description:** Allows dispatchers to adjust the severity rating of an existing active call if conditions worsen.
- **Inputs / Triggers:** Select Escalation Option → Enter Call ID → Enter new severity level (1–5).
- **Expected System Action:** Validates active status, updates severity rating, recalculates required personnel (Severity×2), upgrades assigned lead officer rank if necessary, and re-renders active queue.

FR 7: Department-Specific Queue Filtering

- **Description:** Enables filtering the active queue view by specific response department.
- **Inputs / Triggers:** Queue View Sub-Menu → Select Filter Choice (1: All, 2: Police, 3: Fire, 4: Paramedics).
- **Expected System Action:** Filters queue display to show only incidents involving the selected department while maintaining pending-top / completed-bottom sorting.

FR 8: Officer Workload Capacity Auto-Check

- **Description:** Prevents over-allocating calls to officers who have reached maximum active case capacity.
- **Inputs / Triggers:** Triggered automatically during FR 1 officer assignment.

- **Expected System Action:** Verifies officer workload. If an assigned officer has ≥ 3 active ongoing cases, the system automatically re-routes assignment to the next available officer of appropriate rank.

FR 9: Shift Telemetry Export on System Exit

- **Description:** Automatically exports shift metrics to a local file upon quitting the application.
- **Inputs / Triggers:** Main Menu Choice 5 (Exit).
- **Expected System Action:** Compiles shift telemetry (total calls logged, pending vs. completed counts, officer case tallies) into shift_summary.txt and terminates process.

Edge Cases

EC ID	Input / Situation	Expected Behaviour
EC 1	Invalid main menu selection(e.g., 8 or abc)	Print Error: Invalid choice. Select 1, 2, 3, 4, 5 or 6. and re-prompt main menu.
EC 2	Severity level is not an integer or outside range 1–5 (e.g., 0, 6, high)	Print Error: Severity must be an integer from 1 to 5. and re-prompt input.
EC 3	Invalid or unrecognized location code (e.g., Downtown, L15)	Print Error: Invalid location. Select from L01 to L10. and re-prompt input.
EC 4	Invalid department selection or syntax (e.g., 1, 4, 0, abc)	Print Error: Select valid departments (1-3) separated by commas. and re-prompt input.
EC 5	Database record search returns no matching entries	Print [RECORD SYSTEM] No matching criminal or civil records found for query. and return to menu.
EC 6	Operator attempts case completion before unit signals scene arrival	Print Error: Cannot complete call. Unit has not confirmed arrival on scene. and block state update.
EC 7	Operator enters a Call ID that does not exist	Print Error: Active pending Call ID not

EC ID	Input / Situation	Expected Behaviour
	or is already completed	found. and return to main menu.
EC 8	Operator attempts severity escalation on a completed call	Print Error: Cannot escalate severity. CALL-XXX is already completed. and block modification.
EC 9	Blank or whitespace-only inputs for Caller Name or Emergency Description	Print Error: Caller name and emergency description cannot be blank. and re-prompt input.
EC 10	All officers in a department reach maximum active capacity (≥3 active cases)	Print [WARNING] All personnel occupied. Call CALL-XXX logged to priority waitlist. and set flag.
EC 11	Operator attempts to exit application while critical (Severity 5) calls are Pending	Print [WARNING] Critical Severity 5 calls are still active. Confirm exit? (Y/N). If N, return to menu.

Test Cases

Traces To	Inputs / Actions	Expected Output
FR 1 & FR 2	Caller: Jane Doe Emergency: Robbery Depts: 1, 3 Loc: L01 Sev: 4 Radio Completion: Y	Logs CALL-001. Assigns Police (8 personnel) and Paramedics (8 personnel). Radio feed streams unit acceptance and scene arrival. Case status sets to Completed credited to lead officer.
FR 3	Main Menu Choice 2 → Search Query: Jane Doe	Console outputs formatted table displaying Jane Doe's criminal record, warrant status, traffic citations, and past location incident history.
FR 4 & FR 5	Select Main Menu Choice 4 (View Queue) after completing CALL-001	Incident Queue displays CALL-001 at the bottom with status Completed . Officer Telemetry table updates assigned lead officer's completed count to 1.

Traces To	Inputs / Actions	Expected Output
FR 6	Select Escalation Option → Call ID: CALL-001 (Severity 2) → New Severity: 5	System updates severity to 5, recalculates required personnel from 4 to 10 (Personnel=Severity×2), upgrades assigned lead officer rank, and re-renders active queue.
FR 7	Queue View → Select Department Filter: 2 (Fire Department)	Queue updates to display only incidents assigned to Fire Department, preserving pending-top / completed-bottom sorting.
FR 8	Log new call when assigned lead officer (Officer Miller) already has 3 active cases	System detects capacity limit (≥3 active calls) and automatically re-routes lead officer assignment to next available qualified officer (Officer Davis).
FR 9	Select Main Menu Choice 5 (Exit)	Console outputs Generating shift_summary.txt... Shift log saved successfully. and terminates process.
EC 1	Main menu selection input 8 or abc	Error: Invalid choice. Select 1, 2, 3, 4, or 5.
EC 2	Severity level input 6 or high	Error: Severity must be an integer from 1 to 5.
EC 3	Location code input L99 or Downtown	Error: Invalid location. Select from L01 to L10.
EC 4	Department multi-selection input 1, 5 or 0	Error: Select valid departments (1-3) separated by commas.
EC 5	Main Menu Choice 2 → Search Query: Unknown Person	[RECORD SYSTEM] No matching criminal or civil records found for query.
EC 6	Complete case while unit status is En Route (before reaching scene)	Error: Cannot complete call. Unit has not confirmed arrival on scene.
EC 7	Enter Call ID CALL-999 in Officer Station	Error: Active pending Call ID not found.

Traces To	Inputs / Actions	Expected Output
EC 8	Escalate severity on CALL-001 when status is already Completed	Error: Cannot escalate severity. CALL-001 is already completed.
EC 9	Caller Name input: (blank/spaces)	Error: Caller name and emergency description cannot be blank.
EC 10	Log new call when all officers in department reach 3 active cases	[WARNING] All personnel occupied. Call CALL-XXX logged to priority waitlist.
EC 11	Main Menu Choice 5 (Exit) while CALL-002 (Severity 5) is Pending	[WARNING] Critical Severity 5 calls are still active. Confirm exit? (Y/N)